

Automating Pragmatically

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Title: Automating Pragmatically

The online discussions of automating can leave me confused.

- Should you automate through the GUI?
- Should GUI automating be banned?
- **Do** all testers need **to** code? **Is** automating part **of** testing **or not**?
- **Do** we need **to** automate **to** get a job?

In this **short session** Alan will discuss automating **from** a pragmatic and contextual **position** and **share** how he thinks about automating.

We will leave **time for** questions, and Alan **is** happy **to** answer questions **on any** testing **or** development related topic.

Naming - Test Automation

- we don't automate testing
- Automation in Testing
- Tools Support
- Testing vs Checking

Use your own words.

Automation

- verb?
- noun?
- Automate, Automated, Automating
- Automatization
- Automator: Programmer, User of Tool
- Automaton: Script, Configured Tool

Why include "Test?"

- Test Strategy we identify aims
- Test Approach we identify things we can automate to support our aims

Automating is a concept independent of testing. We Automate Stuff To Help Us Test.

Software Development Automation

- CI, CD, make, automated build
- dependency management
- code completion
- IDE, Static Analysis
- Compilers, Assemblers, Interpreters

We automate throughout the Software Development process.

Use your own words

I avoid saying "Test Automation"

- Automated Execution of Path X
- Assertions to cover acceptance criteria

Choose specific phrases for the concept or point.

Who do you believe?

- Should you automate through the GUI?
- Should GUI automating be banned?
- BDD is not Testing

Believe no-one

- look for absolutes
- identify "but..."
- evaluate approaches

Build your own models. Believe your own experiences.

Future Skills

- Do all testers need to code?
- Is automating part of testing or not?
- Do we need to automate to get a job?

Does it matter?

- we work in teams
- what does your current environment need?
- what are you interested in?
- what are your constraints? time, money, skills, opportunities
- any related weak signals? AI, Security, Ops, Performance

Keep Learning

- Build the path that is appropriate for you
- Build on your strengths
- Don't ignore your weaknesses

Be interested in Software Development as a whole.

A Model of Testing

Comparing a thing to a model of a thing.

- Modelling
- Observation
- Reflection
- Interrogation
- Manipulation

Models of Automated Execution

- System as a Graph
- Paths through a System
- Data flows through paths
- Invariant Data Controls Paths
- Variant Data
 - Partitions
 - Abstracted

System as a Graph

- there are many executable models
 - events, state, flow
- model based testing
 - all testing is model based
 - execution can be model driven

Paths through a System

- execution follows a path
- manually
 - via test script
 - exploratory objective
- automated
 - code
 - model traversal
 - keywords
 - declarative implementation (BDD)

Flow abstraction.

Data Flows Through Paths

- Login (username, password),
- Create New User (username, email, role)

Gives us:

- Data Driven Testing
- BDD example tables
- Certain paths require Invariant data (admin role)
- Equivalence Partitions (valid email, invalid email)

From Modelling, comes Coverage

Automate Tactically or Strategically

- Strategically
 - long term,
 - team agreed,
 - maintained,
 - specific tools
- Tactically
 - short term,
 - specific objective,
 - single use/person,
 - adhoc scripts

What will work for you?

- know your outcomes
- you can only automate what you know to be possible
- experiment tactically
- experience refactors to strategic implementation
- responsibility and ownership

About Alan Richardson

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